

Week 1

1. Existing Software Landscape and Competitive Analysis

The parking technology market already has many applications and platforms designed to make parking easier for drivers and parking operators. These solutions provide features like reservations, mobile payments, and parking passes. Major platforms in this market include SpotHero, ParkWhiz, and ParkMobile. Understanding these existing systems is important because the proposed Smart Parking Platform is not intended to simply recreate an existing parking application. Instead, the project will combine successful features from existing platforms with additional real-time parking, analytics, operator, and smart-city capabilities.

SpotHero

Spot hero allows drivers to search for parking, compare locations and prices, reserve spaces, and pay in advance. It also provides parking operators with tools to manage reservations, inventory, availability, rates, revenue, and facility performance. Its strengths include an established parking marketplace, a convenient reservation process, and useful operator tools. However, this creates an opportunity for the proposed platform to place greater emphasis on live physical occupancy, predictive availability, city transportation integration, and real-time information from parking infrastructure.

ParkWhiz

Park Whiz provides a similar parking search and reservation experience by allowing users to search by destination and time, compare nearby garages and lots, review facility information, pay digitally, and receive a parking pass. Its strengths include convenience, digital payments, event parking, and navigation-related integrations. The proposed system can differentiate itself through deeper real-time occupancy monitoring, operator analytics, city-level information, and predictive parking recommendations.

ParkMobile

Park Mobile provides mobile parking payments and parking reservations. Its zone-parking system allows drivers to enter a parking zone, choose their vehicle and parking duration, and pay through the application rather than using a physical parking meter. Its strength is the convenience of mobile payment for on-demand parking, although functionality may vary depending on the participating location. ParkSphere will attempt to create a more consistent experience that combines discovery, real-time availability, reservations, navigation, payments, and facility management.

Overall, the competitive analysis shows that many individual components of the proposed system already exist. Drivers can already reserve parking, pay digitally, use maps, and receive electronic parking passes, while operators can manage inventory, pricing, reservations, and facility information. Therefore, ParkSphere's competitive advantage will not simply come from adding more features. The platform will focus on creating a unified ecosystem connecting drivers, parking operators, parking infrastructure, and city transportation departments through a real-time system. The proposed platform will differentiate itself through accurate real-time availability, occupancy monitoring, predictive analytics, AI-assisted parking recommendations, operator intelligence, and city transportation integration. Where supported, availability information could be collected through parking facility APIs, entrance and exit systems, cameras, license-plate recognition systems, and IoT parking sensors. Historical occupancy, reservations, traffic, nearby events, time of day, and other available information could also be used to forecast parking demand and recommend parking based on availability, distance, price, and travel time.

2. Vision and Scope (may change later)

2.1 Company Introduction and Project Acquisition

ParkSphere is a transportation technology company specializing in smart mobility, location intelligence, payment technology, data analytics, and urban infrastructure software. The company develops large-scale technology solutions connecting consumers, businesses, and transportation infrastructure and employs project managers, business analysts, software engineers, mobile developers, cloud engineers, cybersecurity specialists, data scientists, UX/UI designers, and quality assurance engineers. For this project, ParkSphere has access to essentially unlimited financial and technical resources, making the fixed semester timeline, third-party dependencies, privacy and security requirements, and applicable laws and regulations the project's primary constraints.

ParkSphere was approached by a coalition of parking facility operators and transportation organizations seeking a more advanced way to manage parking and connect drivers with available parking capacity. Operators reported difficulties maintaining accurate occupancy information, optimizing prices, analyzing utilization, and connecting drivers with unused parking spaces. Drivers also experience difficulty knowing where parking is available before arriving at their destination. After evaluating these problems, ParkSphere proposed a Parking Platform connecting drivers, parking operators, payment providers, mapping services, parking infrastructure, and transportation organizations.

2.2 Project Overview, Business Need, and Vision

The Parking Platform will be a mobile and web-based solution through which drivers will be able to find, compare, book, get directions to and pay for parking. The participating parking

facilities will provide the information about parking spaces and facilities to the platform and will allow drivers to use interactive maps to see parking space locations, availability, prices, operating hours, restrictions, and amenities. The management dashboard will enable parking operators to track occupancy rates, reservations, prices, revenue, and utilization rates of facilities.

The problem comes from the large number of hours spent by drivers when looking for parking spots, especially in areas with traffic and near places where many people come. On the other hand, parking operators need to have more efficient solutions to monitor occupancy, pricing, reservations and increase utilization rates of their parking spots. Therefore, the vision is to create the intelligent parking environment that will save drivers' time during their search for parking and allow parking facilities to connect drivers with available parking options. In the long term, the platform should work as an intelligent parking system connecting drivers, parking facilities, parking operators and transportation agencies and not just as a parking reservation app.

2.3 Project Objectives and Initial Scope

Key goals of the project will be to shorten drivers' time spent looking for parking, provide accurate availability data in real-time, enable online reservations and payments, increase occupancy of parking facilities, supply operators with real-time information about occupancy, help make informed pricing decisions, and to supply analytics to optimize parking operations. Integration with mapping/navigation services, payment providers, and selected parking facilities will be provided in a secure manner as part of the project.

The preliminary scope covers mobile applications for iOS and Android, a responsive web application, user registration and authentication, vehicle profiles, parking search, real-time availability, interactive map view, booking of parking facilities, payments via digital wallet, receipts, history of reservations, integration with navigation service, notifications, dashboards for operators, pricing and occupancy management system, reporting and analytics, administration functions, third-party API integration, and suitable security and privacy features. Possible future or extended functionality includes intelligent parking recommendations based on AI, predictive availability and occupancy data, demand forecasting, recommendation for dynamic pricing, parking sensors, license-plate recognition, availability of EV chargers, integration with smart cities, demand prediction for events, and automatic entry/exit from garages.

3. Software Requirements Specification (SRS)

3.1 Purpose and System Users

The purpose of this Software Requirements Specification is to establish the initial functional and nonfunctional requirements for the ParkSphere Parking Platform. Since this is the first SRS draft, the requirements are expected to change as stakeholder analysis, system design, risk analysis, testing requirements, and additional research are completed throughout the semester. Primary

actors include drivers and vehicle owners, parking facility operators, city transportation departments, system administrators, finance and billing personnel, third-party payment providers, mapping/navigation providers, and external parking infrastructure systems.

3.2 Functional Requirements

The system shall allow users to securely register, authenticate, and manage their accounts and vehicles. Drivers shall be able to search for parking by destination and view participating facilities on an interactive map with information including location, price, operating hours, restrictions, amenities, and available parking inventory. The system shall allow eligible parking inventory to be reserved and shall verify availability before confirming a reservation. The platform shall integrate with approved third-party payment providers to process digital payments and generate reservation confirmations and receipts. Users shall be able to view current and previous reservations, transaction information, and notifications concerning reservations, payments, cancellations, and parking expiration.

The platform shall integrate with external mapping and navigation services to provide directions to selected parking facilities. Parking operators shall have access to a secure management dashboard where they can manage facilities, parking inventory, pricing, operating information, and reservations. The system shall provide occupancy, utilization, reservation, and revenue reporting. Authorized administrators shall also be able to manage users, operators, facilities, permissions, integrations, and platform configurations.

3.3 Nonfunctional Requirements

The platform shall protect sensitive user and payment information through appropriate security controls and encryption, with operator and administrative functions restricted according to user roles. The system shall be designed for high availability and scalability so that increasing numbers of drivers, parking facilities, and transactions can be supported without significant performance degradation. Real-time parking information shall be updated as quickly as technically possible based on information received from facilities and external systems. The mobile and web applications shall provide accessible and easy-to-use interfaces and comply with applicable privacy requirements, payment security requirements, city ordinances, and other relevant laws. Third-party integrations shall also be designed so that the failure of an external service does not unnecessarily cause failure of the entire platform.

4. Initial Use Cases

These use cases represent the first version of the system requirements and will be expanded and refined throughout the semester.

ID	Use Case	Primary Actor	Description
UC-01	Create Account	Driver (not a user yet)	Creates a new platform account.
UC-02	Login In	User	Securely authenticates into the platform.
UC-03	Manage Account	User	Updates personal and account information.
UC-04	Manage Vehicle	Driver	Adds, edits, or removes vehicle information.
UC-05	Search for Parking	Driver	Searches for parking near a destination.
UC-06	View Interactive Parking Map	Driver	Views of participating parking facilities on a map.
UC-07	View Real-Time Availability	Driver	Views current availability reported by a participating facility.
UC-08	View Facility Details	Driver	Reviews price, hours, restrictions, amenities, and facility information.
UC-09	Reserve Parking	Driver	Reserves available parking for a specified period.
UC-10	Pay for Parking	Driver	Completes a digital parking payment.
UC-11	Cancel Reservation	Driver	Cancels an eligible parking reservation.
UC-12	View Reservation History	Driver	Reviews current and previous reservations.
UC-13	Access Receipt	Driver	Views or downloads a transaction receipt.

UC-14	Navigate to Parking	Driver	Opens navigation to the selected parking facility.
UC-15	Receive Notification	Driver	Receives reservation, payment, expiration, and other parking alerts.
UC-16	Manage Parking Facility	operator	Updates facility information, hours, restrictions, and amenities.
UC-17	Manage Parking Inventory	operator	Controls reservable parking inventory and availability.
UC-18	Manage Pricing	operator	Establishes and modifies parking prices.
UC-19	Monitor Occupancy and Analytics	operator	Reviews occupancy, utilization, reservations, revenue, and performance.
UC-20	Administer Platform	operator	Manages users, facilities, operators, integrations, permissions, and settings.

4.1 Detailed Use Case Example, UC-09: Reserve Parking

Primary Actor: Driver

objective : Reserve parking at a selected facility for a specified date and time.

Preconditions met: The driver has selected a participating parking facility, reservable inventory is available, and the required account, vehicle, and payment information has been provided.

Trigger: The driver selects the option to reserve parking.

Intended flow: The system displays the selected facility, reservation period, price, and applicable conditions. The driver reviews the information and selects a payment method. Before processing the transaction, the system verifies that parking inventory remains available. The

payment provider authorizes the transaction, the system creates the reservation, and the appropriate parking inventory is updated. A reservation confirmation and receipt are then generated for the driver, and the reservation becomes available to the parking operator.

Alternative Flow: If parking inventory is no longer available before the transaction is completed, the system prevents the reservation and provides alternative parking options. If payment authorization fails, the reservation is not completed and the driver can select another payment method.